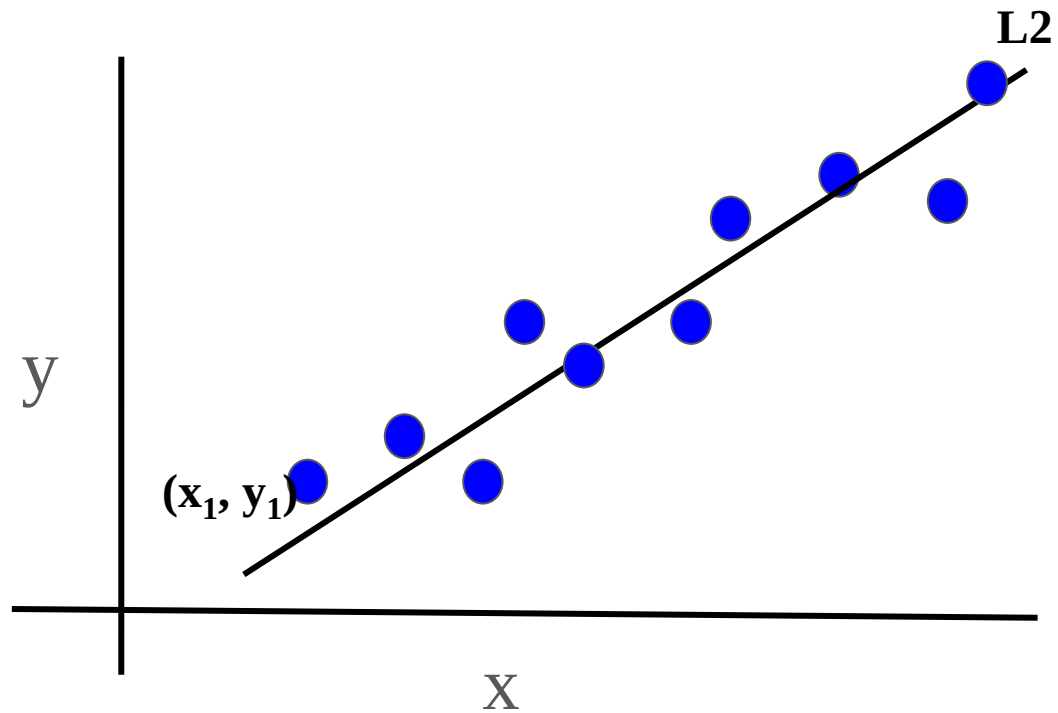


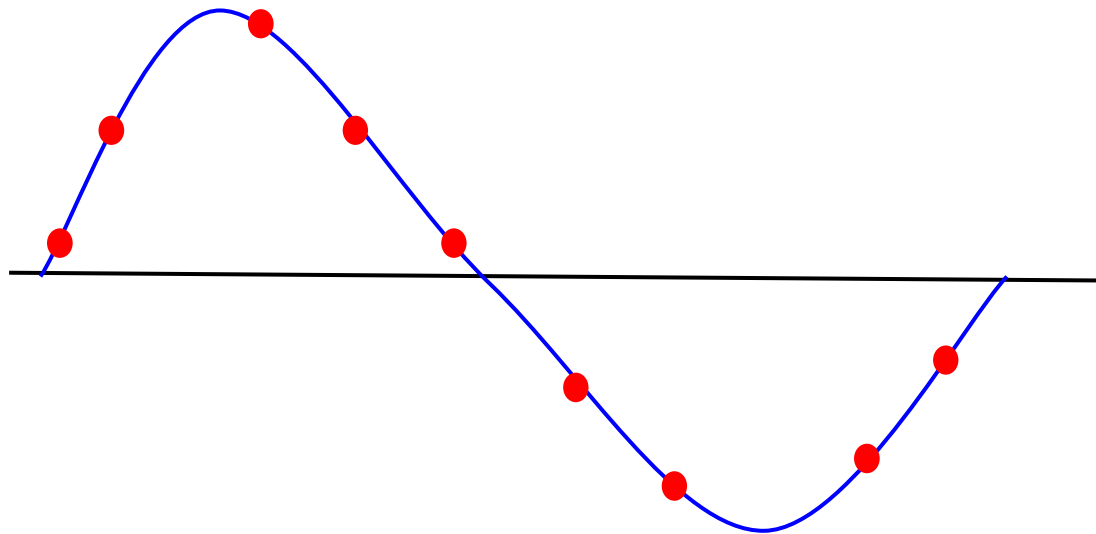
Linear Regression 2

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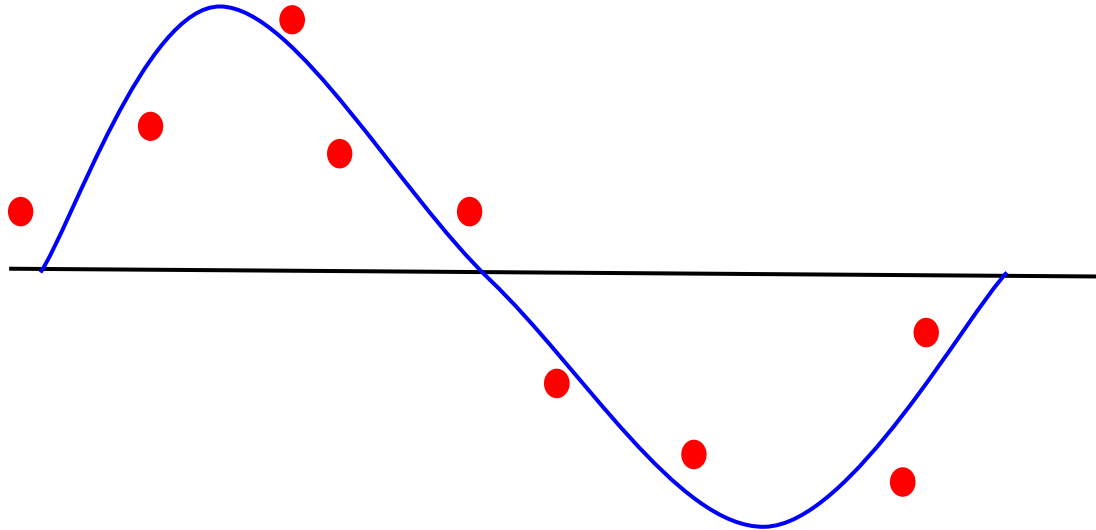


Best line L2

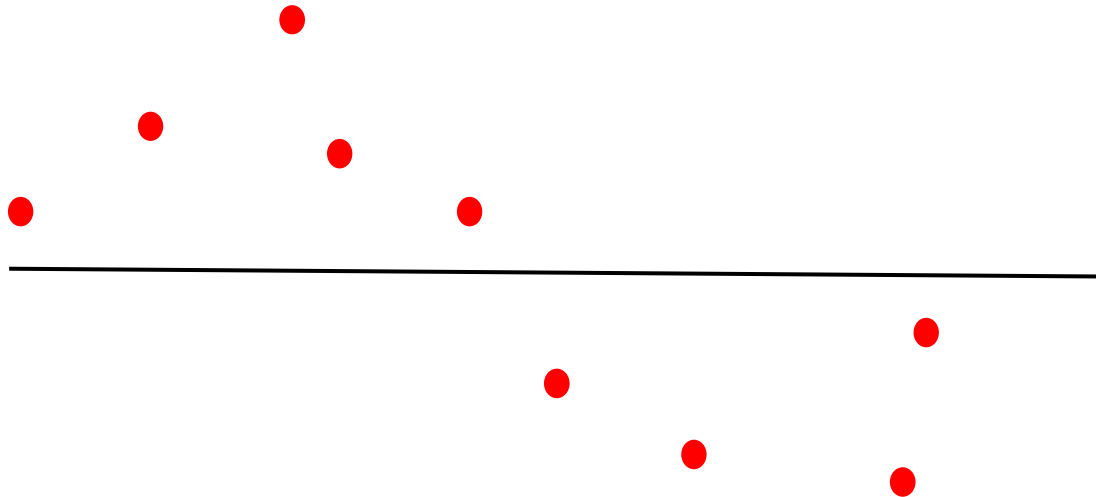
Can we fit only straight lines?



Noise

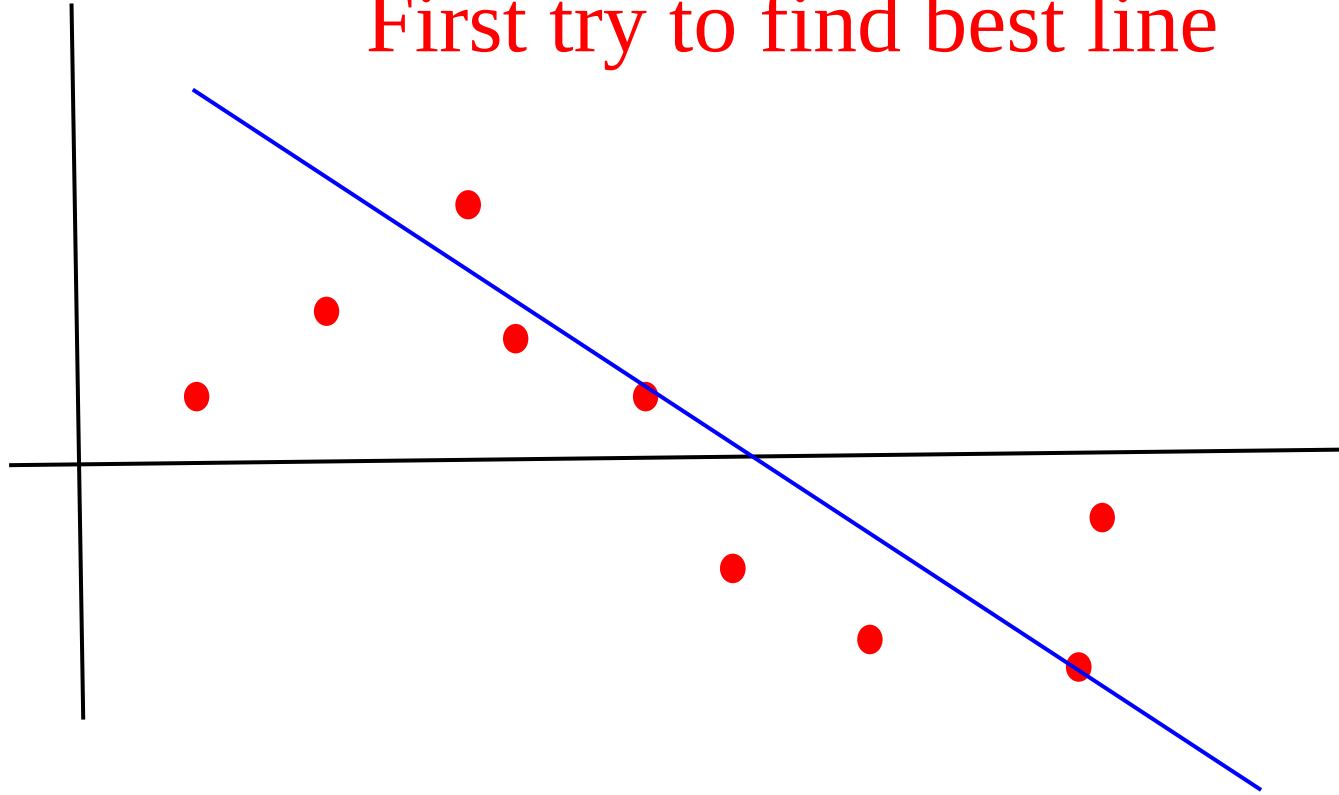


Given dataset only



Task to find the true function

First try to find best line



Line Equation

$$m * x + c$$

Alternatively,

$$w_1 * x + w_0$$

What is the degree of equation of line?

1(One)

Can we fit degree 2 polynomial?

How degree 2 polynomial look like?

$$w_0 + w_1 * x + w_2 * x^2$$

Let's apply old tricks from previous
lecture

Dataset: x, y

We want to find

w_0, w_1, w_2 in equation

$$w_0 + w_1 * x + w_2 * x^2$$

We need to preprocess the data to fit the

$$\text{model: } w_0 + w_1 * x + w_2 * x^2$$

Preprocessing

$$x_1 \Rightarrow x_1, x_1^2$$

$$x_2 \Rightarrow x_2, x_2^2$$

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• • •

$$x_{10} \Rightarrow x_{10}, x_{10}^2$$

Arrange the preprocessed data into matrix

$$\begin{bmatrix} x_1 & x_1^2 \\ x_2 & x_2^2 \\ \cdot & \cdot \\ \cdot & \cdot \\ x_{10} & x_{10}^2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \\ \cdot \\ \cdot \\ y_{10} \end{bmatrix}$$

Compute the prediction

$$p_1 = w_0 + w_1 * x_1 + w_2 * x_1^2$$

$$p_2 = w_0 + w_1 * x_2 + w_2 * x_2^2$$

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• • • •

$$p_{10} = w_0 + w_1 * x_{10} + w_2 * x_{10}^2$$

We can arrange it as

$$\begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ 1 & x_{10} & x_{10}^2 \end{bmatrix}$$

$$\begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix}$$

So, we can rewrite the predictions as follows:

$$\begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_{10} \end{bmatrix} = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_{10} & x_{10}^2 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix}$$

P**X****W**

How to compute error?

Remember last lecture

$$\text{Total error} = (\mathbf{y}_1 - \mathbf{p}_1)^2 + (\mathbf{y}_2 - \mathbf{p}_2)^2 + \dots + (\mathbf{y}_{10} - \mathbf{p}_{10})^2$$

Rewrite error function using linear algorithm

$$\text{Total error} = \mathbf{e}_1^2 + \mathbf{e}_2^2 + \dots + \mathbf{e}_{10}^2$$

Where, $\mathbf{e}_i = y_i - p_i$

Total error can be written vector multiplication

$$\begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \cdots & \mathbf{e}_{10} \end{bmatrix} \begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \\ \cdot \\ \mathbf{e}_{10} \end{bmatrix}$$

$\mathbf{e}^T$

$\mathbf{e}$

$$\text{Total error} = \mathbf{e}_1^2 + \mathbf{e}_2^2 + \cdots + \mathbf{e}_{10}^2 = \mathbf{e}^T \mathbf{e}$$

$$P = X W$$

$$\text{error} = P - Y$$

Or,

$$\text{Error} = XW - Y$$

Now, we can minimise the error using derivative method and find the optimal polynomial for the given dataset.

$$\nabla_W (XW - Y)^T (XW - Y) = 0$$

$$W = (X^T X)^{-1} X^T Y$$

Code

[https://github.com/aruntakhur/GEU_ML_2024/blob/
main/LinearRegressionLab1GEU2024.ipynb](https://github.com/aruntakhur/GEU_ML_2024/blob/main/LinearRegressionLab1GEU2024.ipynb)

Thanks